initial.txt

```
START 100
MOVER AREG A
ADD BREG B
MOVEM AREG C
A DS 1
B DC 5
C DS 1
END
```

```
import java.io.*;
import java.util.*;

public class Main {

    public static void main(String[] args) {

        int address = 0, scount = 0, lcount = 0;

        String symbols[] = new String[10];
        String literals[] = new String[10];
        int sadd[] = new int[10];
        int ladd[] = new int[10];

        String IS[] = {"ADD", "MOVEM", "MOVER"};
        String REG[] = {"AREG", "BREG", "CREG", "DREG"};

        try {
            BufferedReader br = new BufferedReader(new FileReader("initial.txt"));

            PrintWriter im = new PrintWriter("IM.txt");
            PrintWriter st = new PrintWriter("ST.txt");
            PrintWriter lt = new PrintWriter("LT.txt");

            String line;

            while ((line = br.readLine()) != null) {

                StringTokenizer tokens = new StringTokenizer(line);

                while (tokens.hasMoreTokens()) {

                    String t = tokens.nextToken();
```

```
if (t.equals("START")) {
    im.print("AD1 ");
}
else if (t.equals("END")) {
    im.print("AD2 ");
}
else if (t.matches("\\d+")) {
    address = Integer.parseInt(t);
    im.println(address);
}
else if (t.equals("ADD") || t.equals("MOVEM") || t.equals("MOVER")) {
    int i = Arrays.asList(IS).indexOf(t);
    im.print("IS" + (i + 1) + " ");
}
else if (Arrays.asList(REG).contains(t)) {
    im.print((Arrays.asList(REG).indexOf(t) + 1) + " ");
}
else if (t.equals("DC")) {
    im.print("DL1 ");
}
else if (t.equals("DS")) {
    im.print("DL2 ");
    int a = Integer.parseInt(tokens.nextToken());
    address += a - 1;
}
else if (t.startsWith("=")) {
    im.print("L" + lcount + " ");

    literals[lcount] = t;
    ladd[lcount] = address;
    lcount++;
}
else if (t.length() == 1) {

    int index = Arrays.asList(symbols).indexOf(t);

    if (index == -1) {
        symbols[scount] = t;
        sadd[scount] = address;
        index = scount;
        scount++;
    }
```

```
                    im.print("S" + index + " ");
                }

            }

            im.println();
            address++;
        }

        for (int i = 0; i < scount; i++) {
            st.println(i + "\t" + symbols[i] + "\t" + sadd[i]);
        }

        for (int i = 0; i < lcount; i++) {
            lt.println(i + "\t" + literals[i] + "\t" + ladd[i]);
        }

        br.close();
        im.close();
        st.close();
        lt.close();

        System.out.println("Pass 1 completed successfully!");

    }
    catch (Exception e) {
        System.out.println(e);
    }
  }
}
```